⇒ ENTROPY

↑

{ measure the Randomness in your data } → $Y \to Y_1 \; Y_2 \; Y_3 \; ---- \; Y_K$

↑

$P(Y_i X)$

$(0,1)$

$$\underline{H(Y)} = - \sum_{i=1}^{K} p(Y_i) \cdot \log_2 \left(P(Y_i) \right) \quad \text{where } \boxed{b=2}$$

'information'

$$= -\underbrace{\frac{9}{14} \log\left(\frac{9}{14}\right)}_{\text{Yes}} - \underbrace{\frac{5}{14} \log\left(\frac{5}{14}\right)}_{\text{No}} \; - - - -$$

$$H(Y) = 0.94$$

Blackboard

1 / 3

$Y \rightarrow$ Yes, No

(case i)

$$D \rightarrow \begin{cases} \text{Yes} \rightarrow 99\% \\ \text{No} \rightarrow 1\% \end{cases}$$

$$H(Y) = -0.99\log(0.99) - 0.01\log(0.01)$$

$$H(Y) = \underline{\underline{0.0801}}$$

(case ii)

$$D \rightarrow \begin{cases} \text{Yes} \rightarrow 50\% \\ \text{No} \rightarrow 50\% \end{cases}$$

$$H(Y) = -0.5\log(0.5) - 0.5\log(0.5)$$

$$\boxed{H(Y) = 1}$$

2 / 3

Case iii

D → Yes → 0%

D → No → 100%

$0 \times \underline{\log(0)} - 1 \times \log(1) \rightarrow 0$

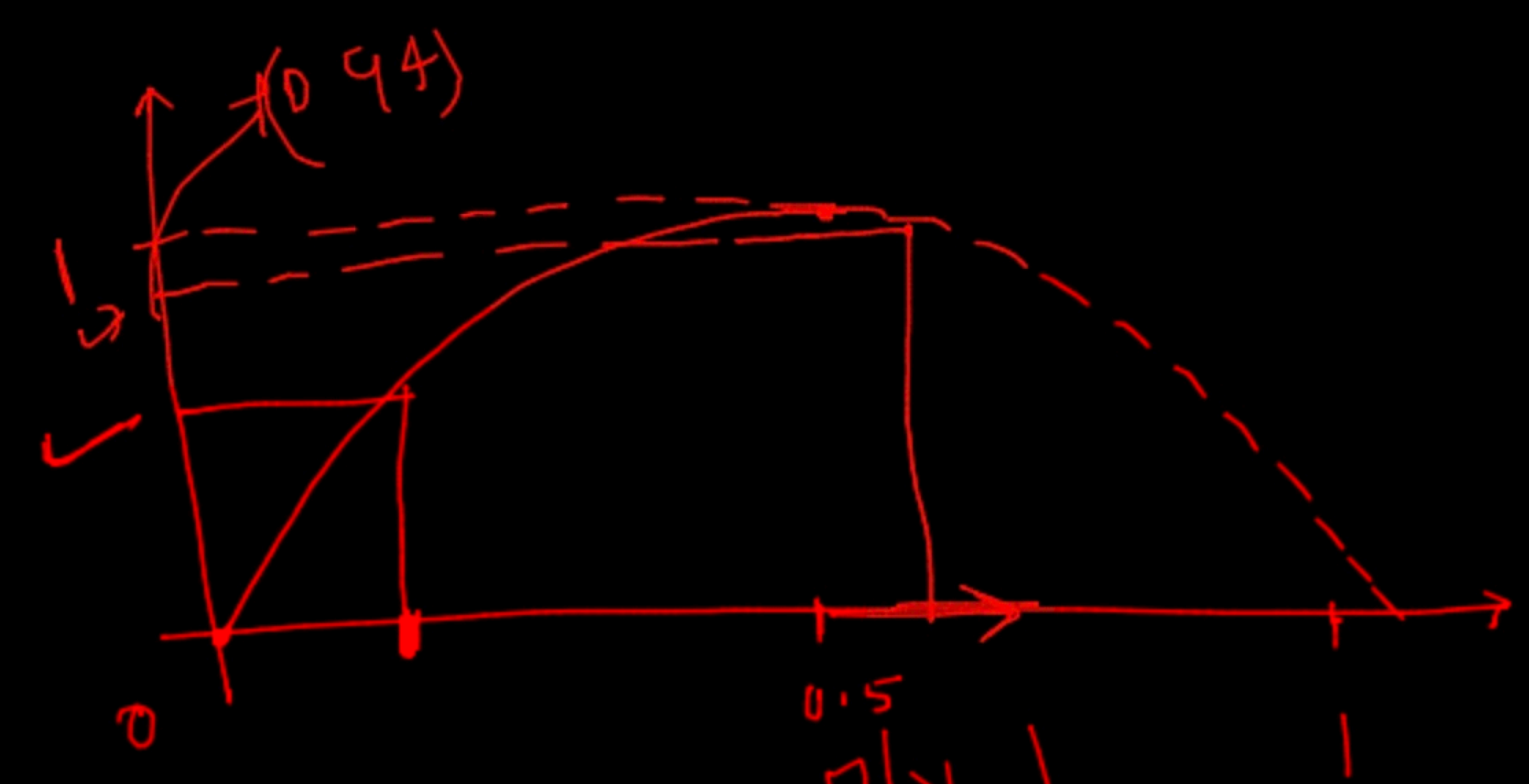
$H(Y) = 0$

(i) ENTROPY — 1 Equally dist

↳ $\underline{\min}$.

$\left(\frac{9}{14}\right)$

(Entropy)



$P(Y_+)$

↑

Yes

$P(Y_+) \Rightarrow 0.2$

$P(Y_-) = 0.8$

① Information gain

Pure Node

{Dataset (Tennis)

$H(Y) \rightarrow \underline{0.94}$

14

Outlook

Sunny?	Rain	Overcast
D1	D2	(D3)
$\lvert D1 \rvert = 5$	$\lvert D_2 \rvert = 5$	$\lvert D \rvert = 4$
$H(D1) = \underline{0.97}$	$H(D2) = 0.97$	$H(D_3) = 0$

Y

IG

Weight ENTROPY

$$= \frac{5}{14} * 0.97 + \frac{5}{14} * 0.97 + \frac{4}{14} \times 0$$

$$\frac{5}{14} * 0.97 + \frac{5}{14} \times 0.97$$

$$\Downarrow$$

$$\cancel{2} * \frac{5}{\cancel{14}_{7}} * 0.97$$

$$\boxed{\text{weight ENTROPY} = \frac{5}{7} * 0.97}$$

IG(Y, outlook)

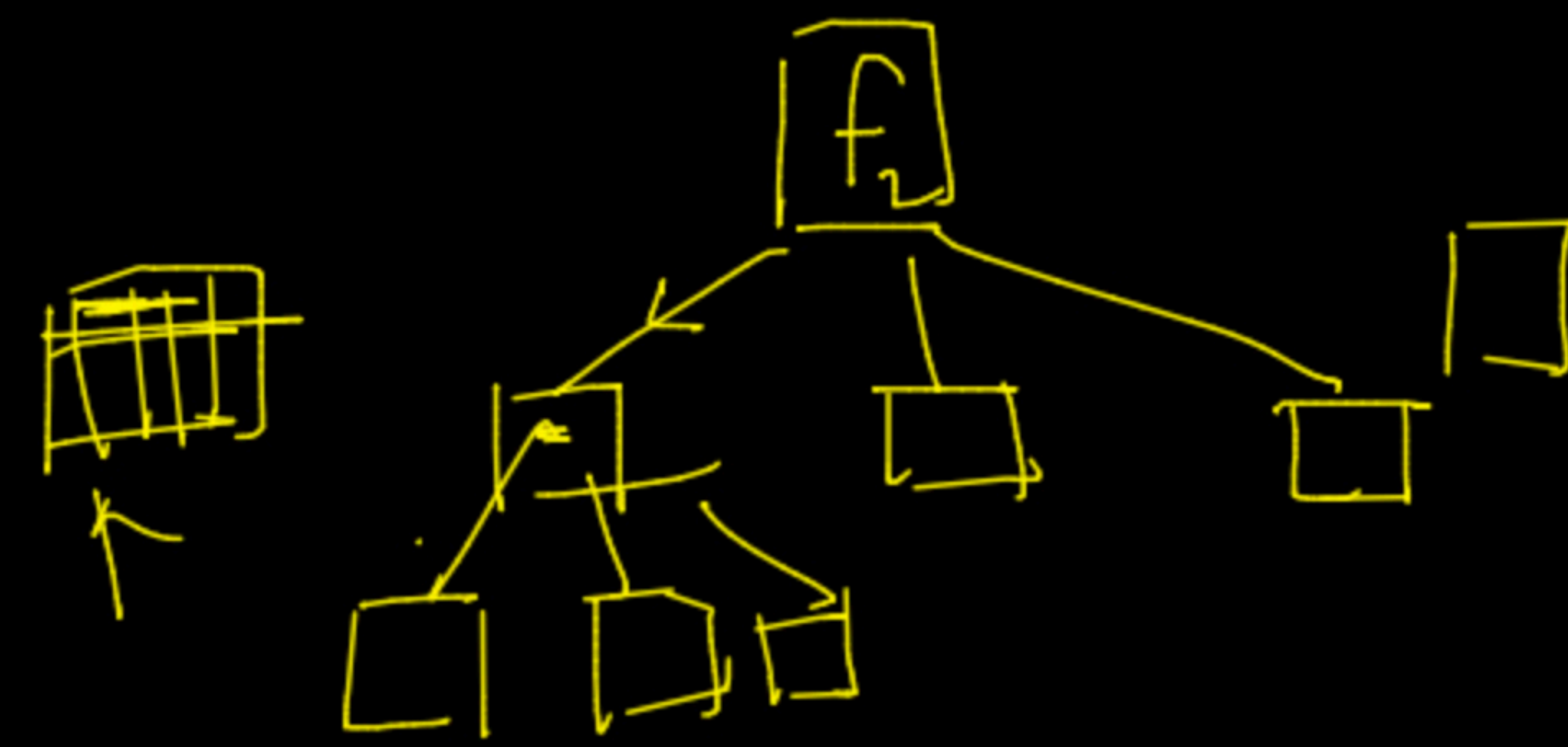
= (weighted entropy after breaking dataset (D, D2, D3)) − (Entropy Before breaking the data)

$$IG(Y, \text{outlo}) = \underbrace{\frac{5}{7} * 0.97}_{} - 0.94$$

IG(Y, outlook)

5 / 5

$feature = [f_1, f_2, f_3, f_4]$



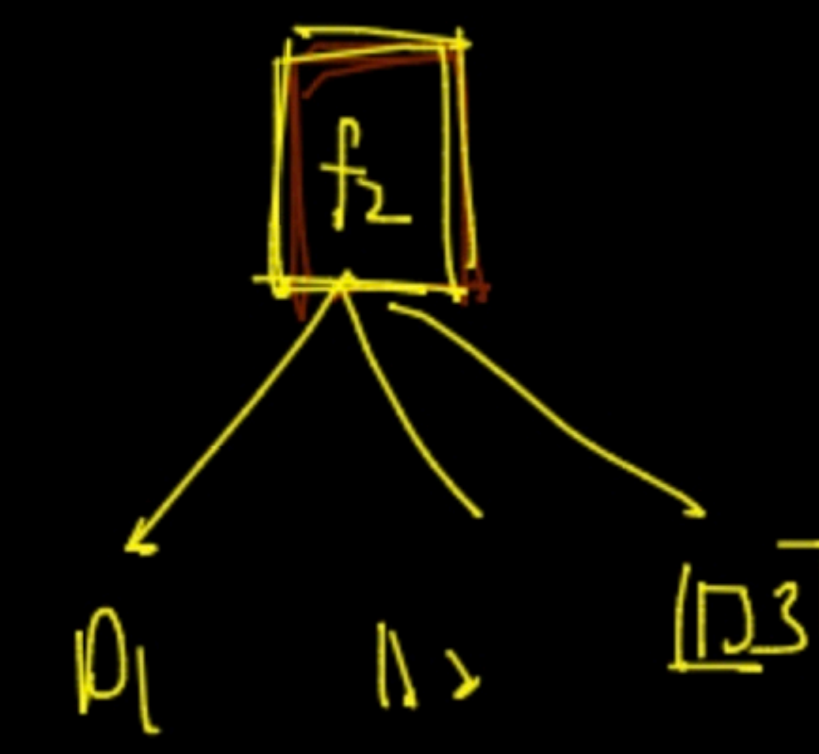
for i in feature

$ig = f\, i$

$f_1 \rightarrow$ $IG(Y, f1) = 0.91$

$IG(Y, f_2) = 0.92$

Parent Node



$IG(Y, f_3) = 0.68$

$IG(Y, f_4) = 0.51$

ONE HOT → train

.fit(X-train[-])

(save it)

at time of predict

[— - - - -]

transform